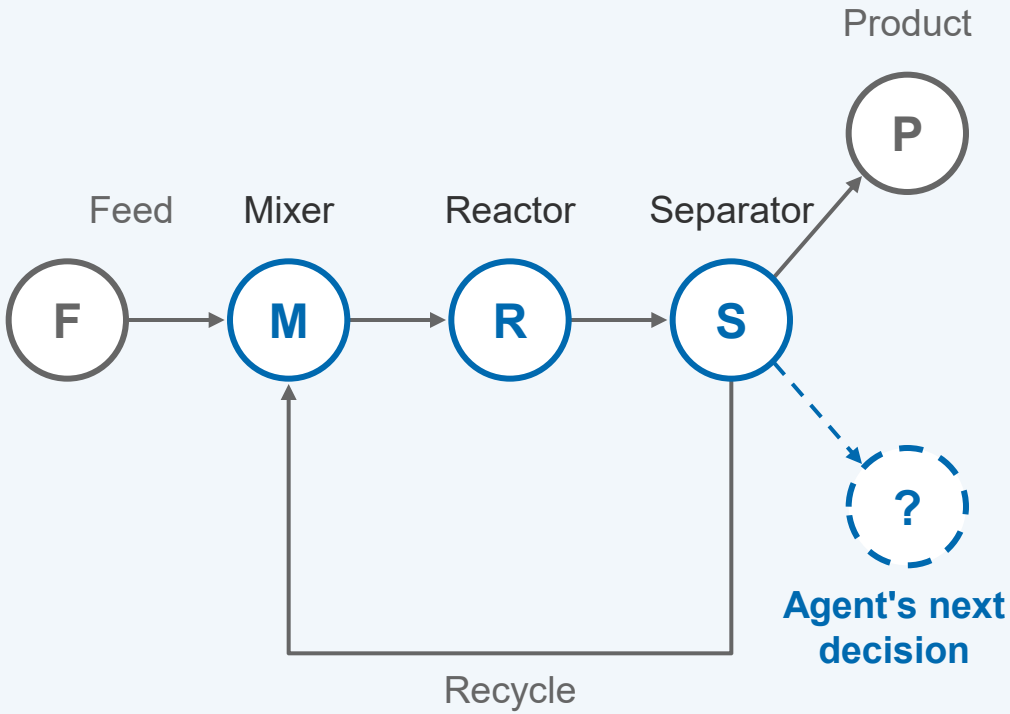
PROCESS SYSTEMS ENGINEERING · RL

RL-PSE

Automated flowsheet generation with graph neural networks and hierarchical reinforcement learning

Gabriel Talero

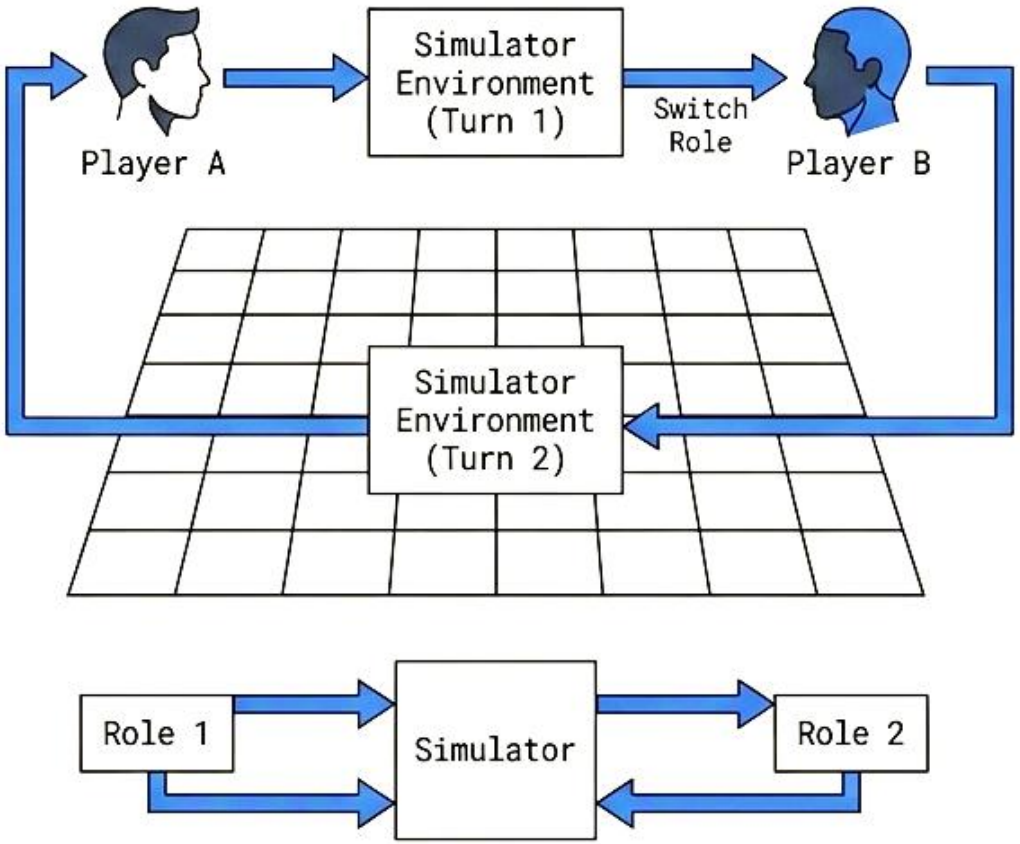
September 2026



Architectural Paradigm Shift: Stops vs. Burger

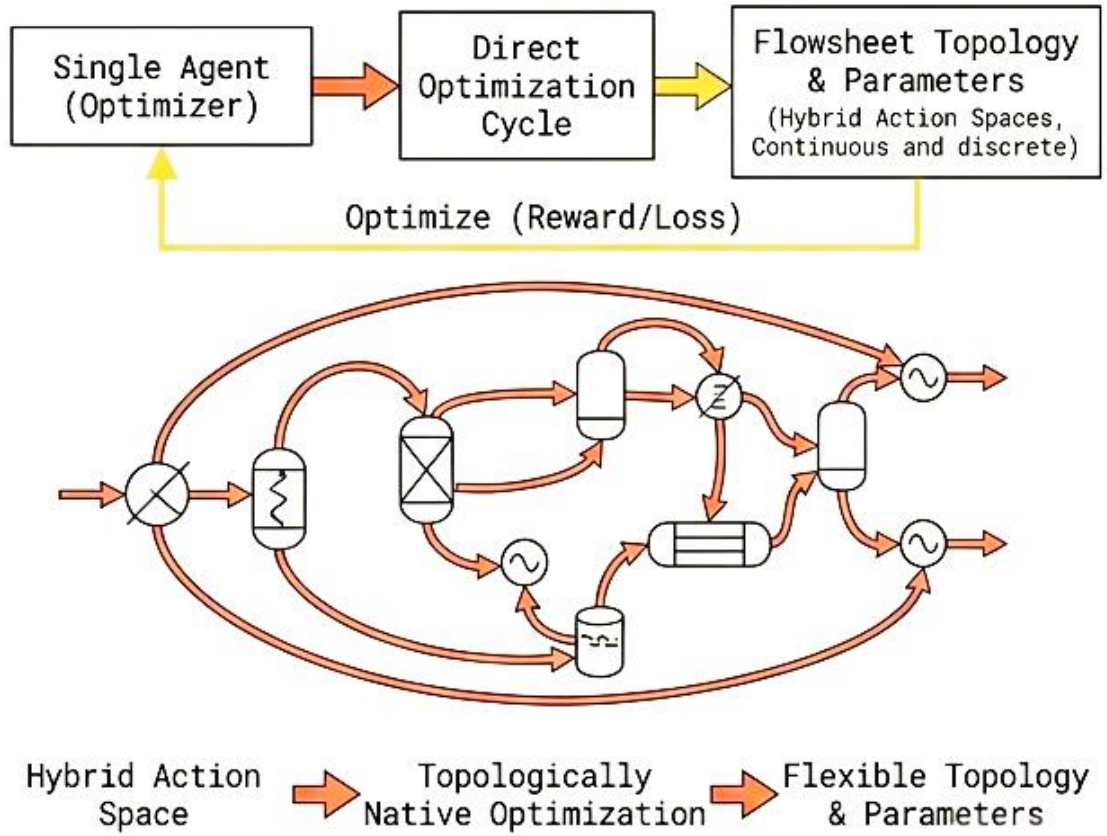
Burger et al. / SynGameZero

Metaphor: The Game Board

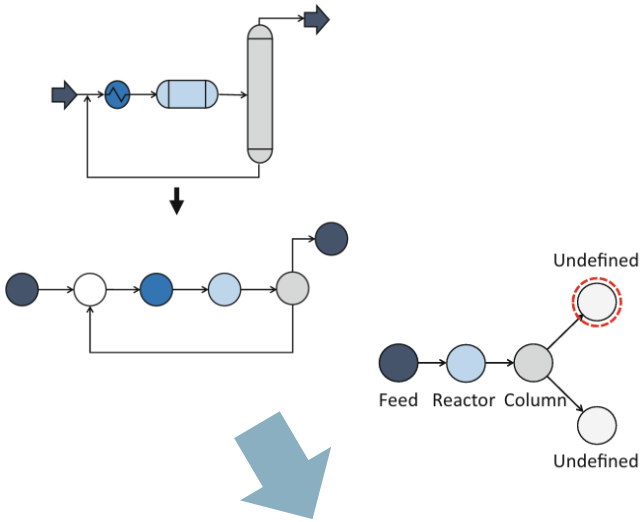


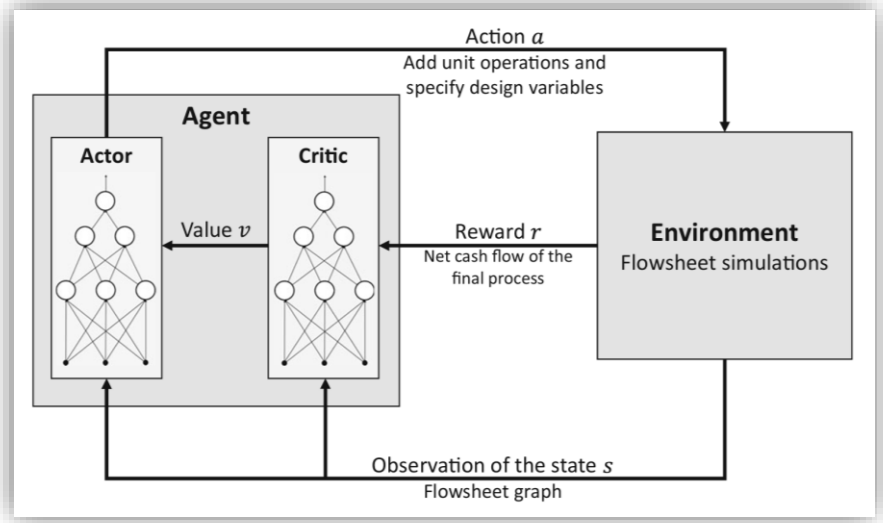
Stops et al. / Hierarchical GNN

Metaphor: The Organic Network



Demo RL-PSE



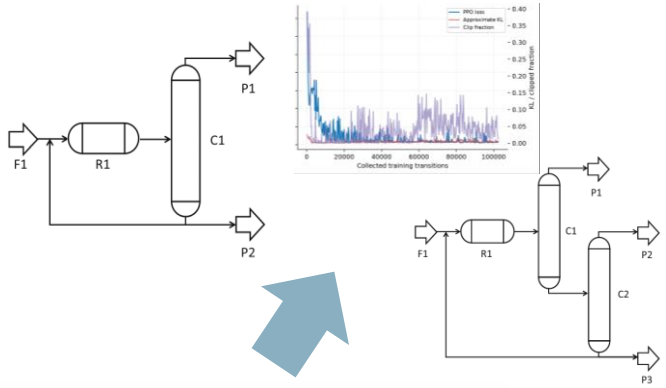


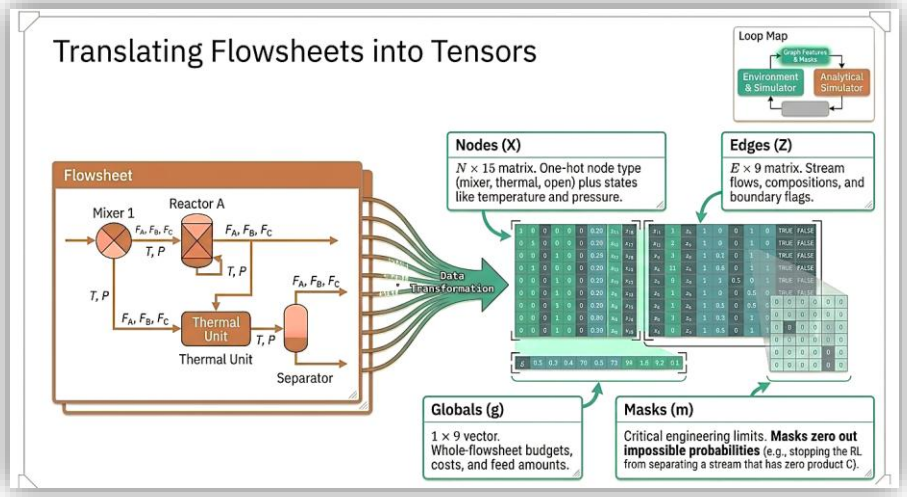
Source: Stops et al. DOI:10.1002/aic.17938

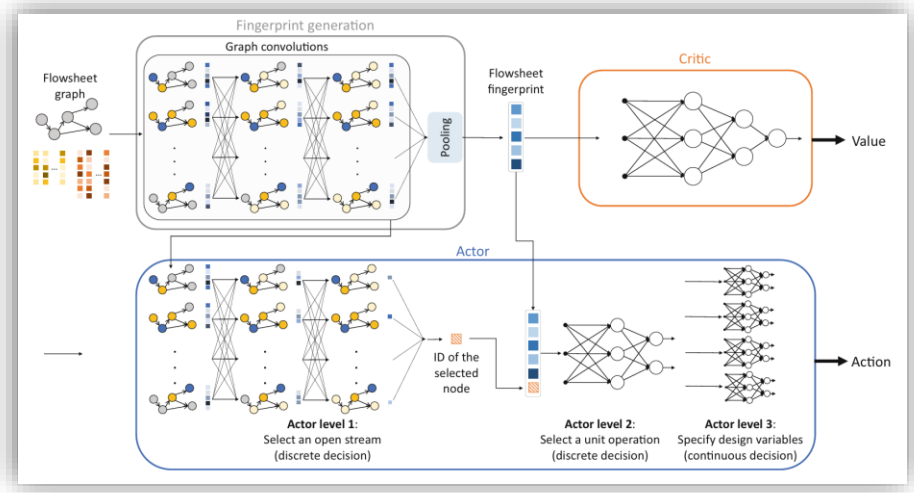
RESEARCH ARTICLE
Process Systems Engineering
AIChE JOURNAL

Flowsheet generation through hierarchical reinforcement learning and graph neural networks

Laura Stops | Roel Leenhouts | Qinghe Gao | Artur M. Schweidtmann

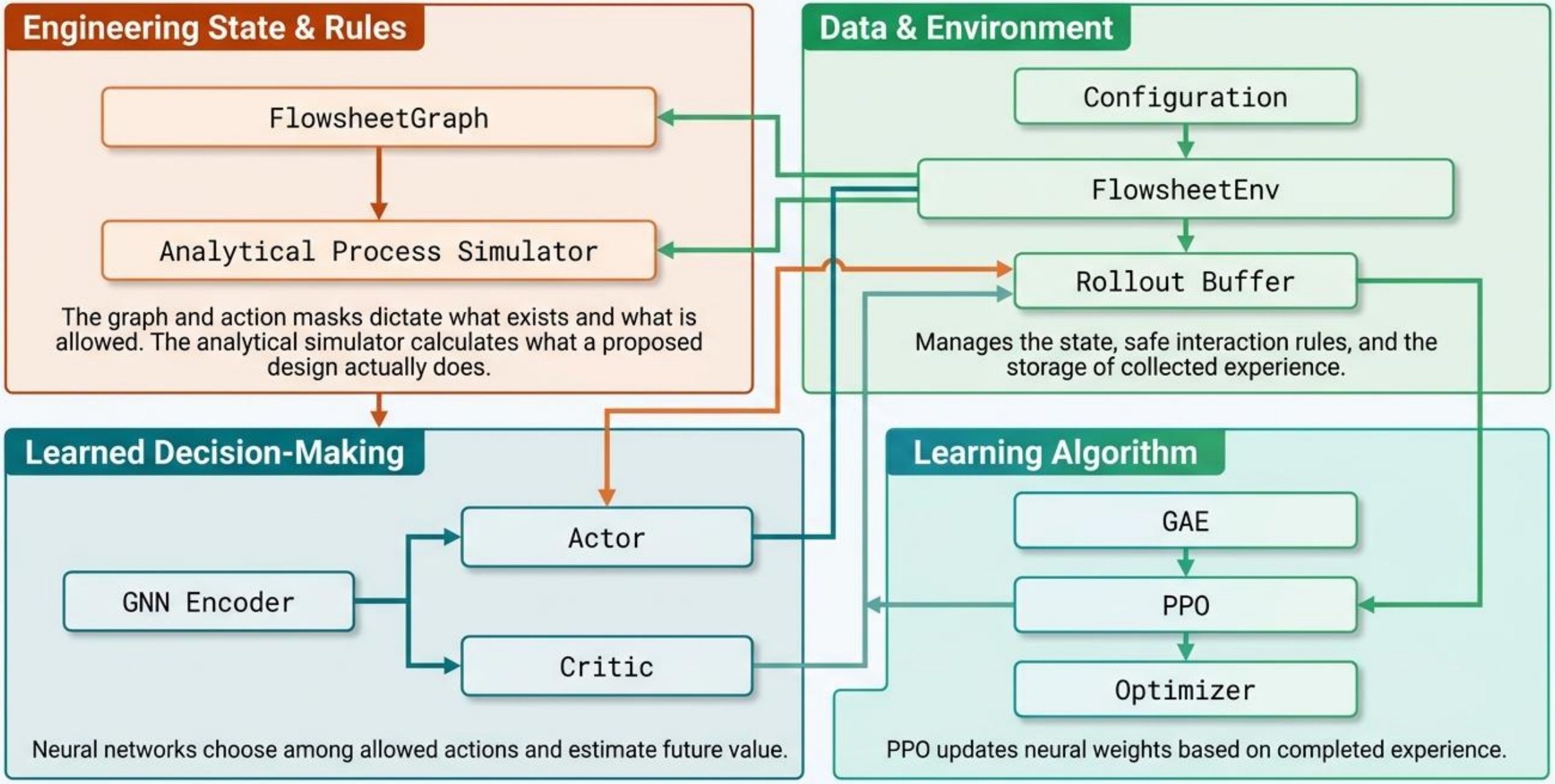






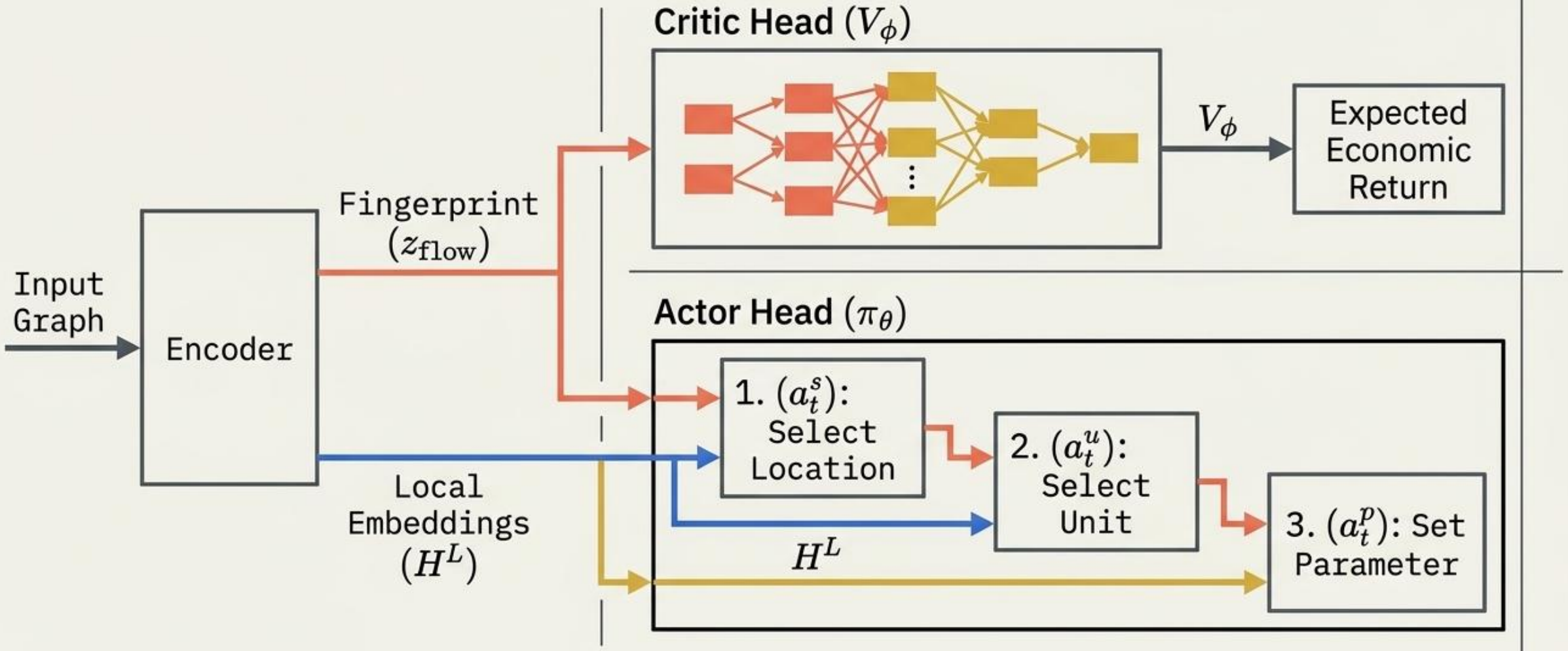
A minimal Python prototype that converts partial flowsheets into graph tensors, encodes them with a GNN, and uses a hierarchical actor–critic agent to select the next stream, unit operation and design variables.

The Complete Workflow Architecture



Gemini Notebook

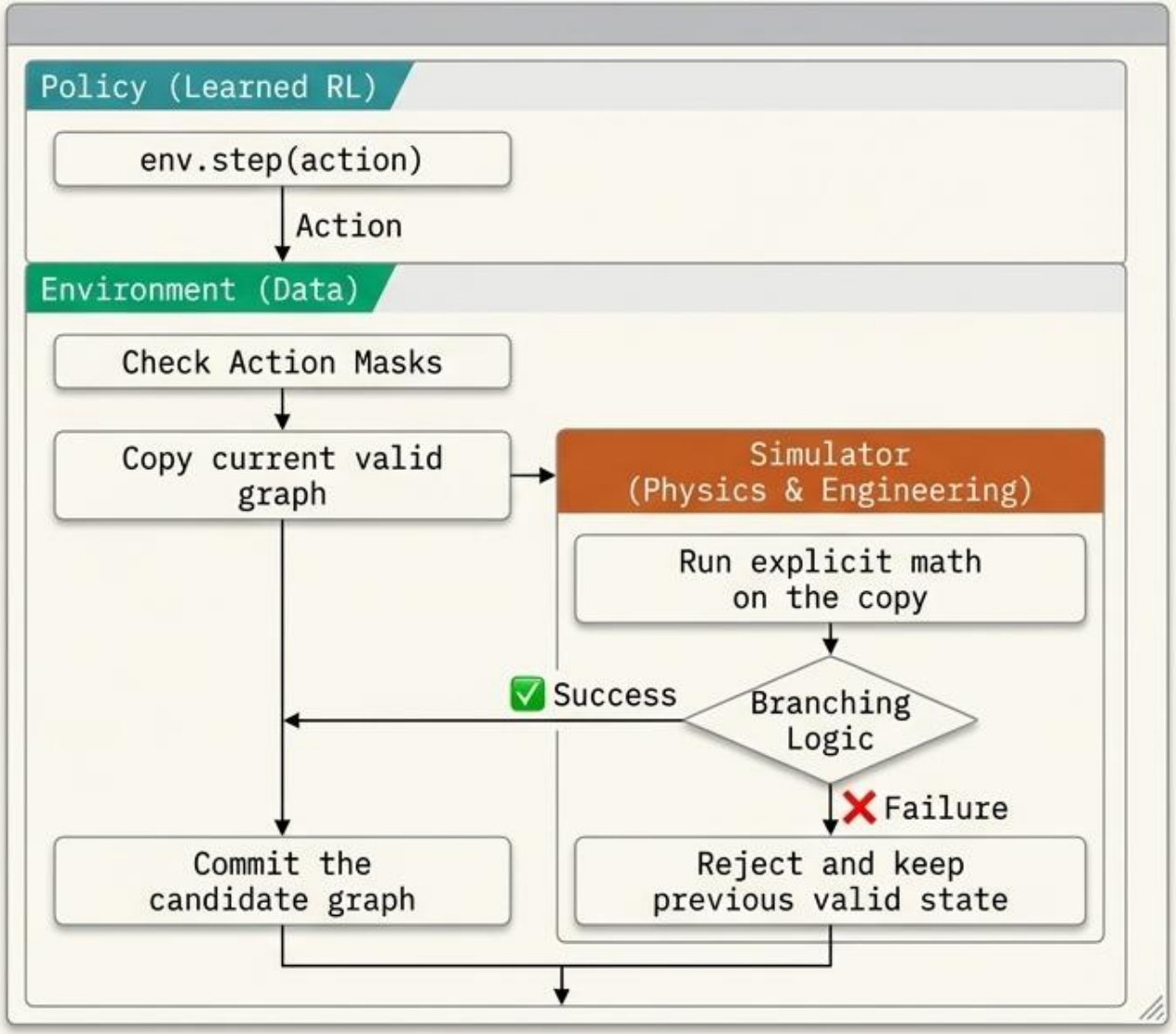
PPO for Hybrid Action Spaces

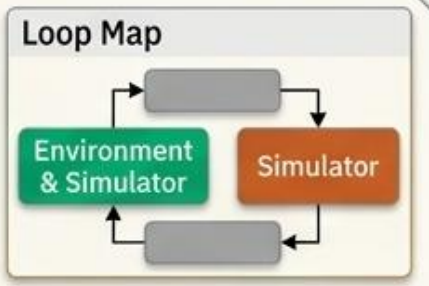


Unlike classical temporally extended 'options', this architecture decomposes a single graph-edit action into dependent sub-actions evaluated sequentially within one environment step.

Gemini Notebook

Safe Trial and Error in the Environment Step





The Transactional Mechanism

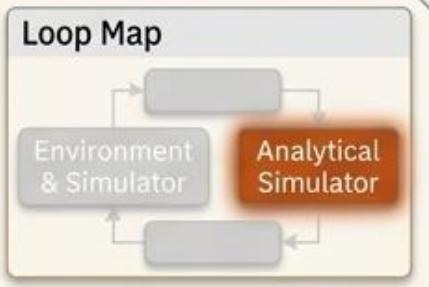
The copy-then-simulate structure ensures a failed physical proposal never corrupts the last mathematically valid graph.

The Rule Baseline

Before training, an unlearned **rule policy** (**mixer → thermal → reactor → separator**) runs purely to prove a feasible construction path actually exists.

Gemini Notebook

Mechanistic Engineering Evaluated Topologically



Mixer

Conserves mass and flow-weights the temperature.

Thermal Unit

Maps a normalized policy action $p \in [0,1]$ to a target temperature:

$$T_{\text{out}} = 280 + 170p$$

Reactor (A+B → C)

Calculates synthetic reaction extent based on kinetics, bounded by available mass:

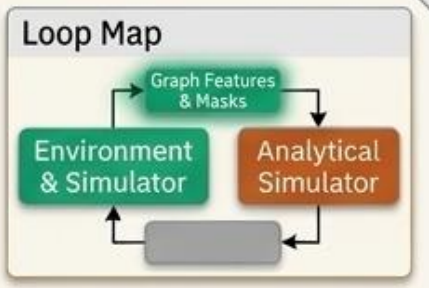
$$\xi = X \min(F_A, F_B)$$

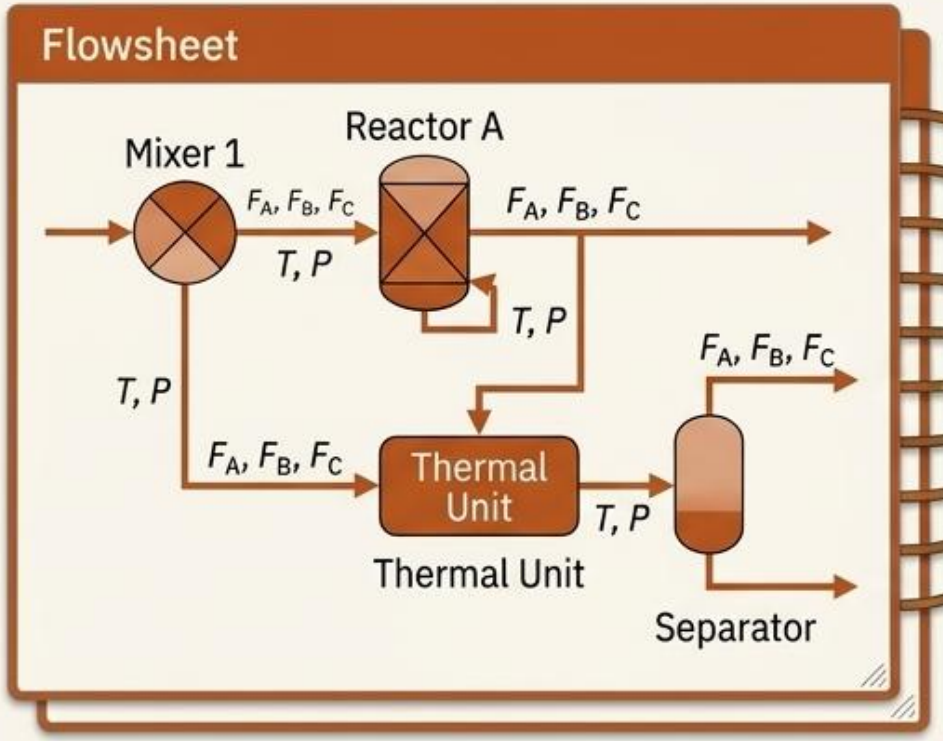
Key Takeaway:

The reaction extent (ξ) **explicitly prevents the consumption of more A or B than physically exists.** This is an **engineered boundary,** not a learned one. The chemistry is just **readable,** mechanistic Python logic.

Gemini Notebook

Translating Flowsheets into Tensors





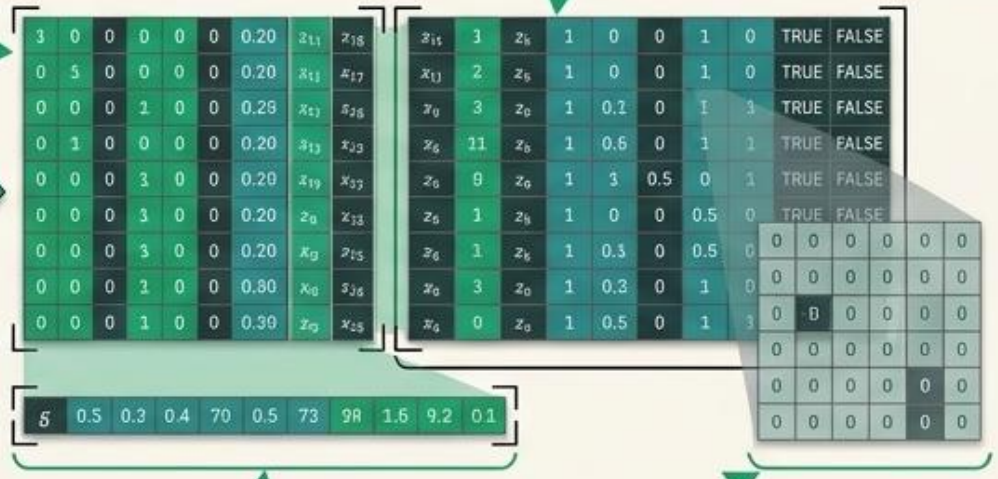
Data Transformation

Nodes (X)

$N \times 15$ matrix. One-hot node type (mixer, thermal, open) plus states like temperature and pressure.

Edges (Z)

$E \times 9$ matrix. Stream flows, compositions, and boundary flags.



Globals (g)

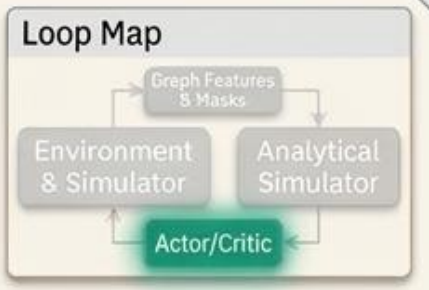
1×9 vector.
Whole-flowsheet budgets, costs, and feed amounts.

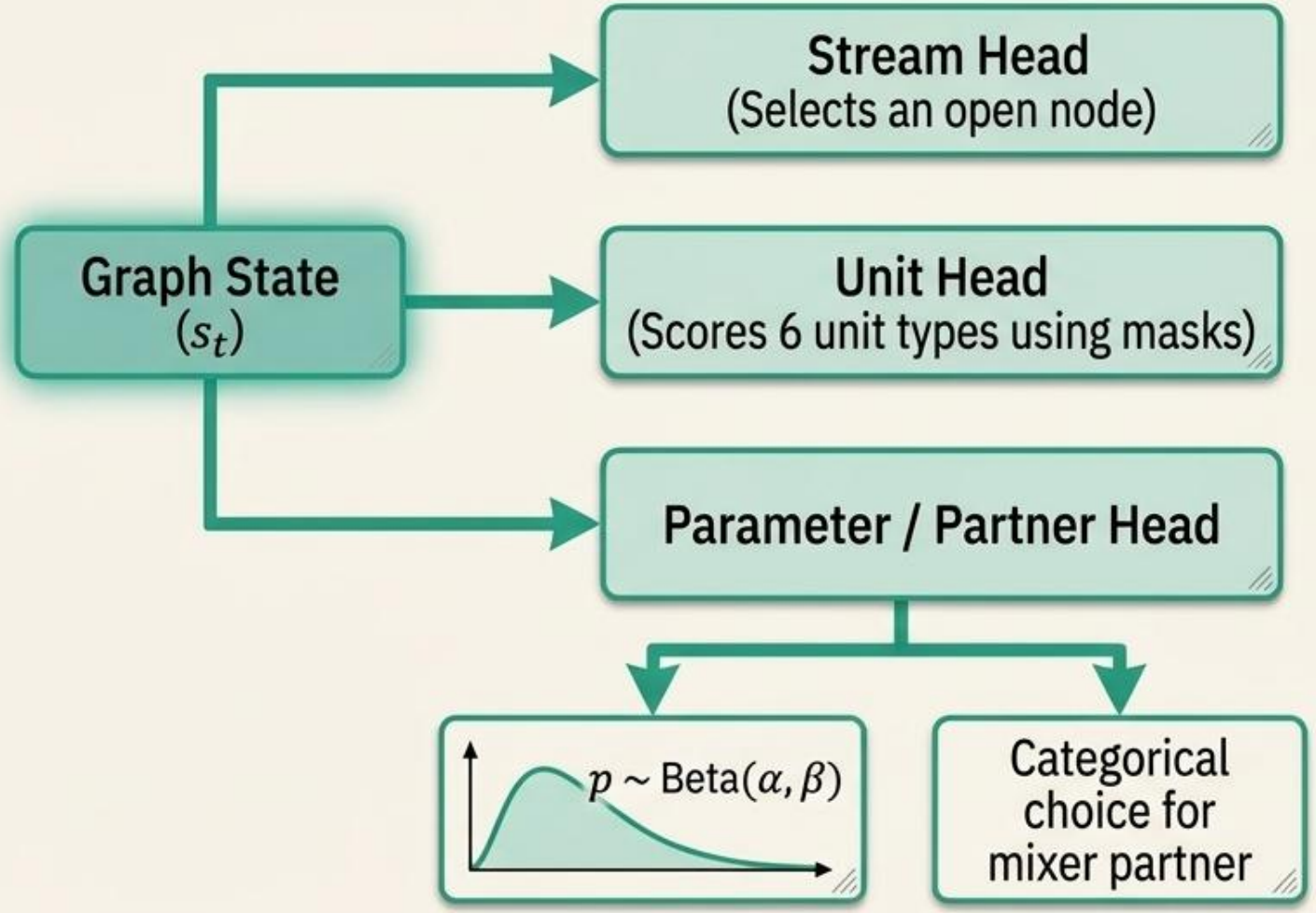
Masks (m)

Critical engineering limits. **Masks zero out impossible probabilities** (e.g., stopping the RL from separating a stream that has zero product C).

Gemini Notebook

Navigating the Hierarchical Actor-Critic





The Actor

The policy factorizes conditionally $(\pi(a_t|s_t))$. Decisions are strictly sequential. If a mixer is chosen, the neural network calculates a partner choice, but ignores continuous parameters.

The Critic

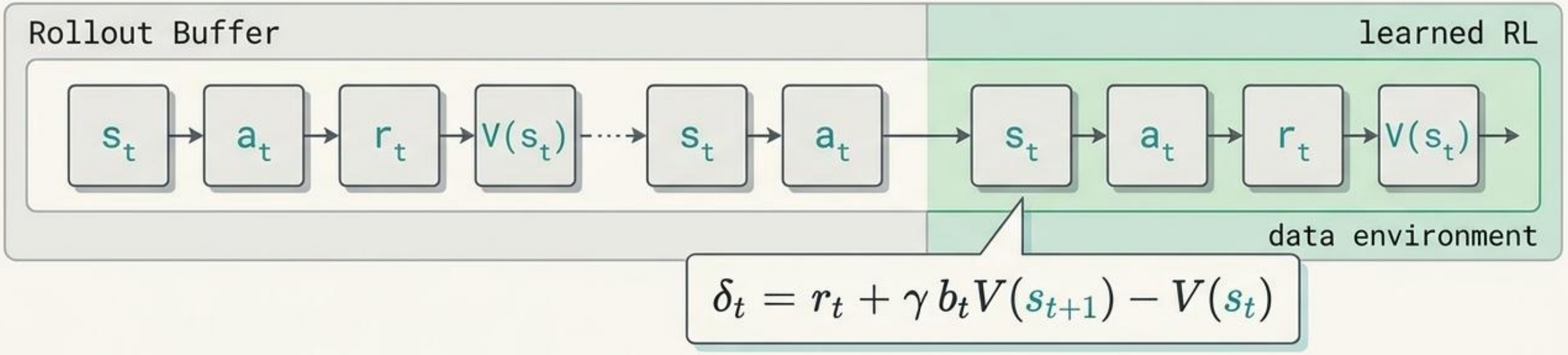
Independently reads the global fingerprint h_G to estimate future discounted reward $V(s_t)$. It predicts value, but does not choose actions.

Gemini Notebook

Judging Experience with Rollouts and GAE

Loop Map

Buffer → GAE



The Collector

The Rollout Collector gathers environment transitions over fixed steps—storing states, chosen actions, actual rewards, and the critic's predicted values before learning occurs.

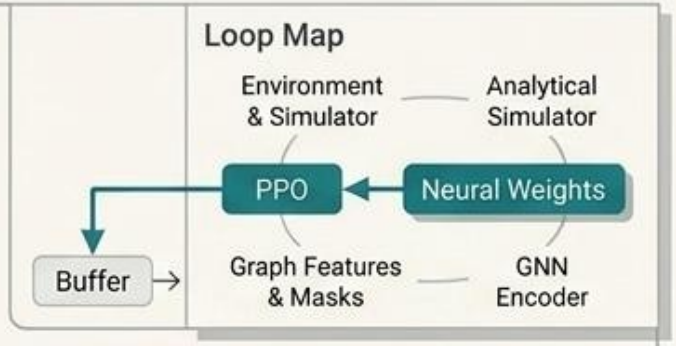
Generalized Advantage Estimation (GAE)

Calculates the “surprise” factor.

- `Advantage > 0`: The sampled design outcome was better than the critic expected.
- `Advantage < 0`: The outcome was worse than expected.

Gemini Notebook

Updating Neural Weights with PPO



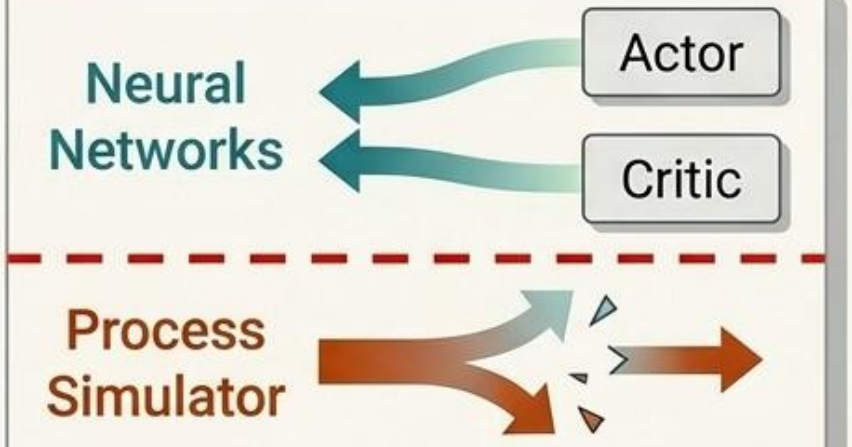
$$\mathcal{L} = \mathcal{L}_{actor} + c_V(\text{Value Loss}) - c_H(\text{Entropy})$$

$\mathcal{L}_{actor}$
trains choices

Value Loss
trains predictions

Entropy
encourages exploration

$\text{clip}(\rho_t, 1-\varepsilon, 1+\varepsilon)$ limits extreme policy updates.

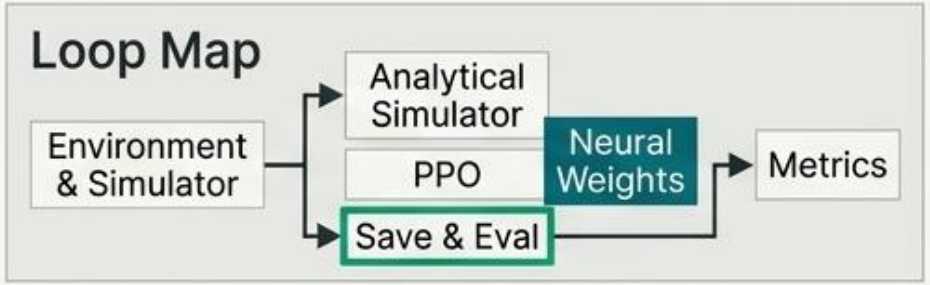


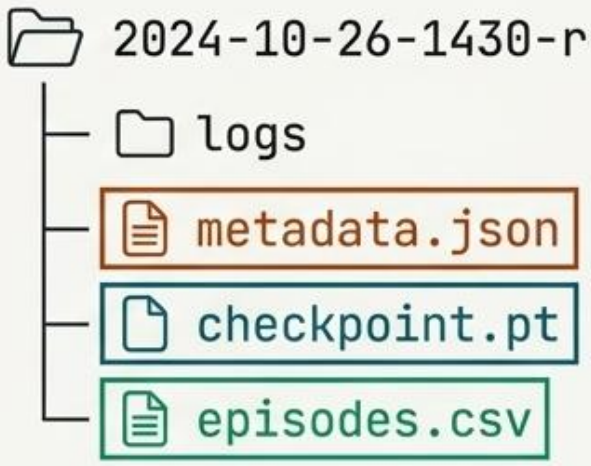
The Gradient Stop

PyTorch auto-differentiation flows backward through the Actor, Critic, and GNN—but stops entirely before the simulator. The mechanistic simulator is never differentiated.

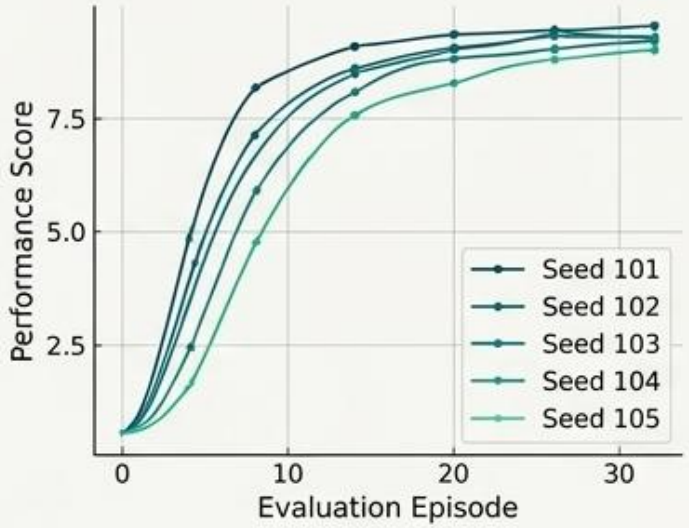
Gemini Notebook

Deterministic Evaluation and Artifacts





Auditability: Every run creates a **sealed folder** with configuration hashes, complete episode logs, and `.pt` checkpoints for exact checkpoints for exact mathematical resumption.

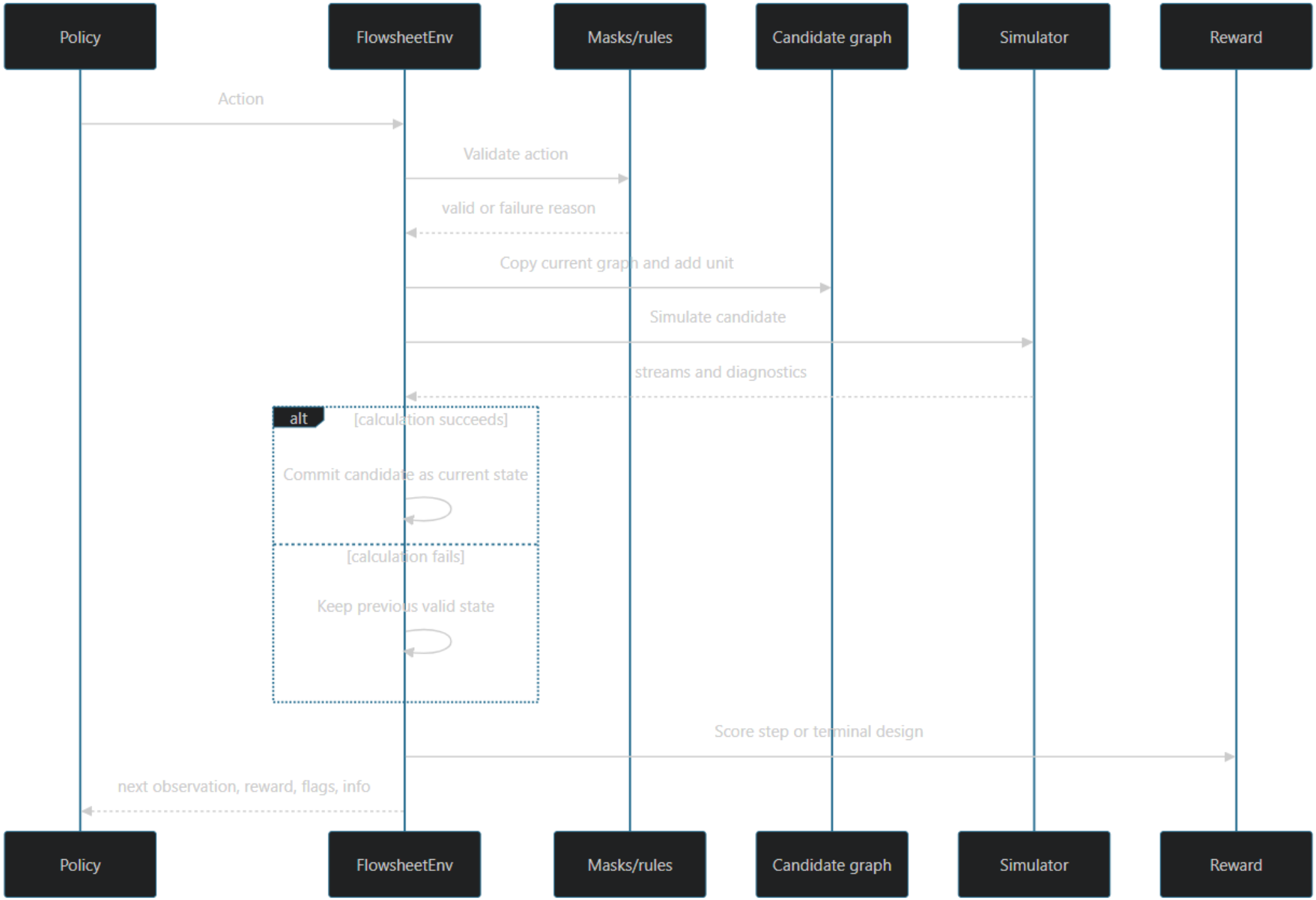


Inference vs. Training
During evaluation, the network uses deterministic categorical `argmax` and `Beta` means. The historical checkpoint's weights do not change.

Clarifying Seeds

Different evaluation feed seeds (e.g., 101–105) test ONE fixed policy under different feed conditions. They are not independent training runs.

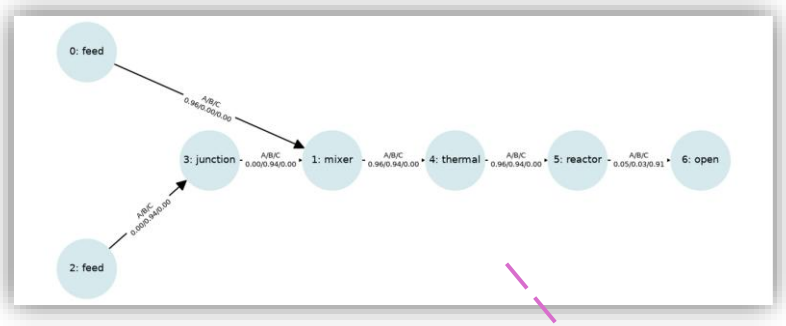
Gemini Notebook

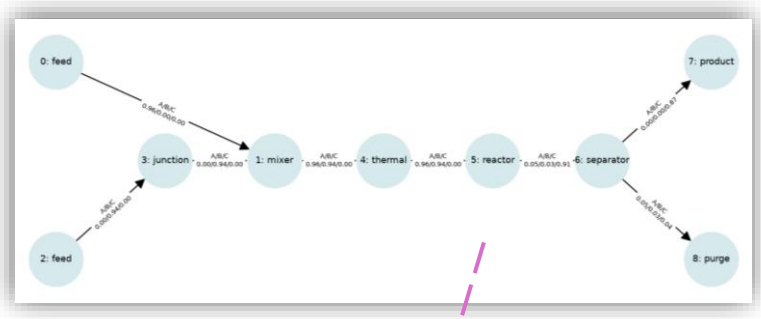

Policy
FlowsheetEnv
Masks/rules
Candidate graph
Simulator
Reward
Action
Validate action
valid or failure reason
Copy current graph and add unit
Simulate candidate
streams and diagnostics
alt
[calculation succeeds]
Commit candidate as current state
[calculation fails]
Keep previous valid state
Score step or terminal design
next observation, reward, flags, info
Policy
FlowsheetEnv
Masks/rules
Candidate graph
Simulator
Reward

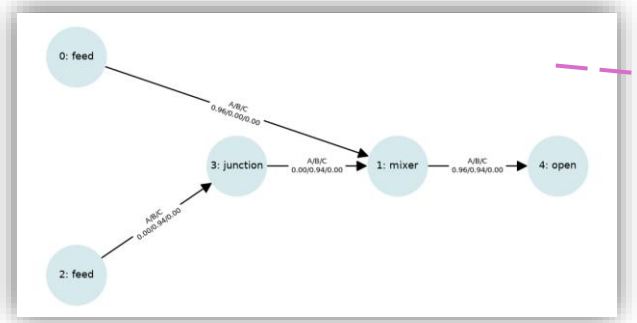
Demo RL-PSE

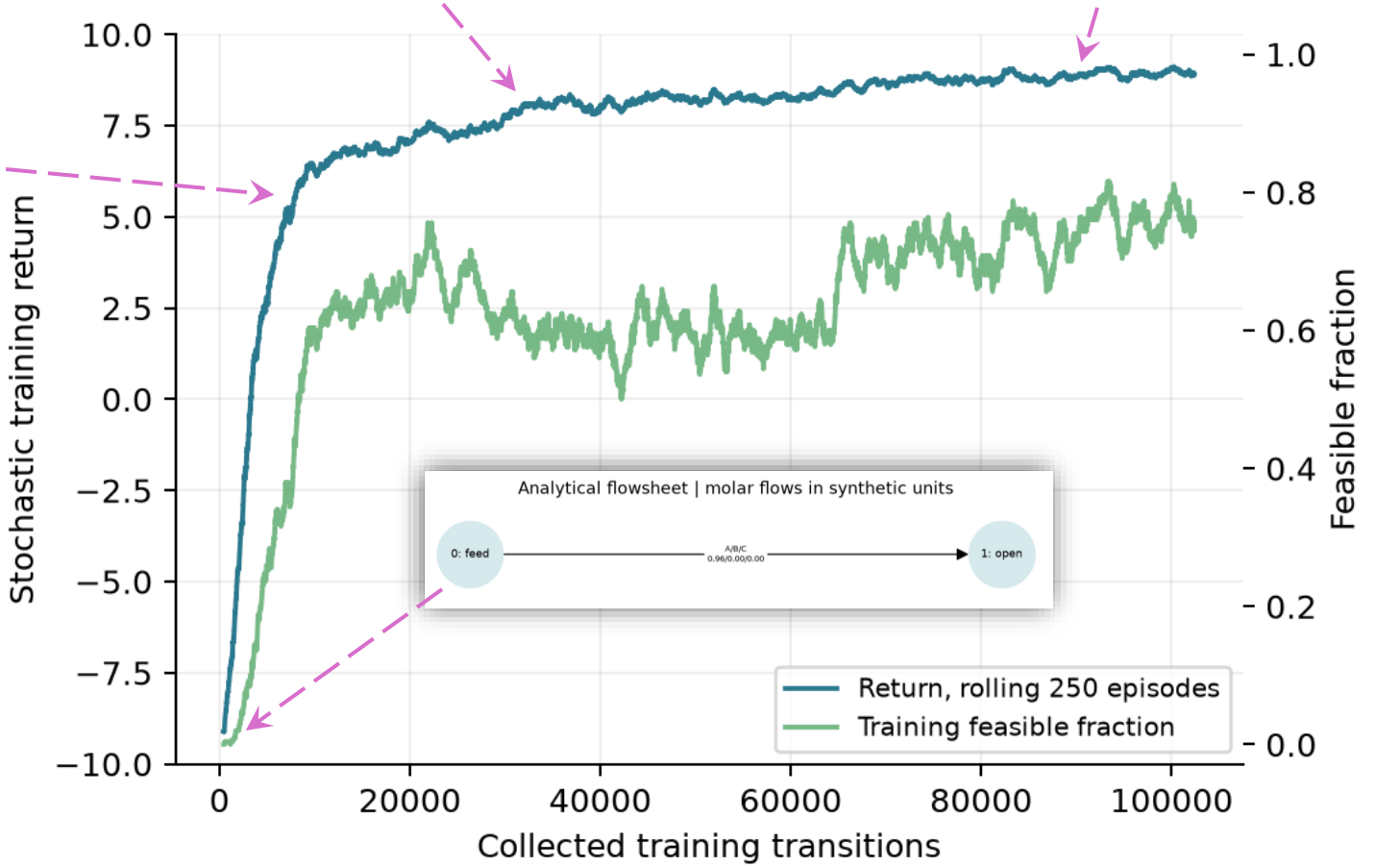
TARGET:

Closed flowsheet; Component and elemental balances satisfied; Product purity ≥ 0.90; Product recovery ≥ 0.60; Prefer lower energy, capital/complexity



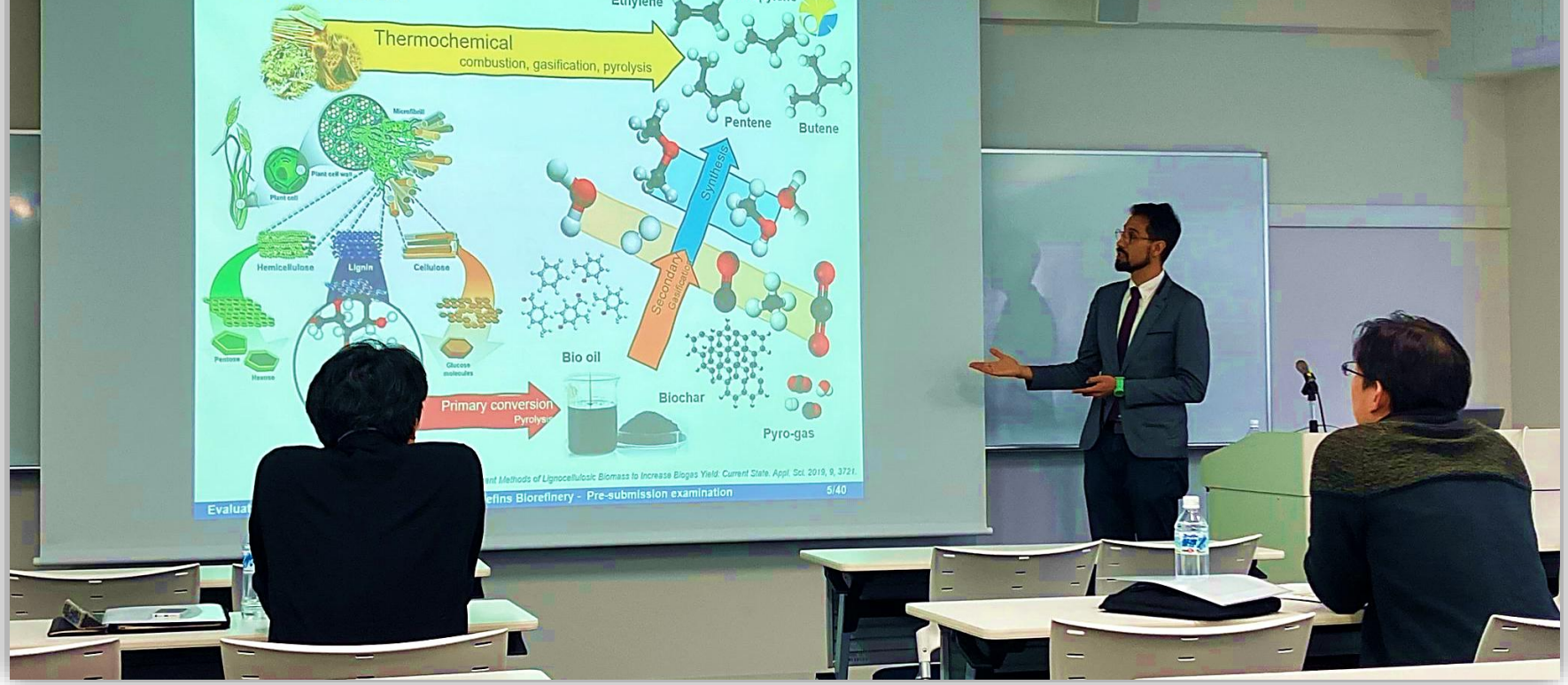






LIMITATIONS:

- **Simplified analytical simulator:** shortcut Python unit models; no rigorous thermodynamics, phase equilibria or Aspen/DWSIM coupling.
- **Synthetic, acyclic scope:** A + B → C system on an acyclic graph; no recycle/dynamic control.
- **No generalisation or global optimality:** case-specific prototype; needs full retraining for new processes.



Thanks for your attention

Questions?

gftaleror@unal.edu.co

linkedin.com/in/gabriel-fernando-talero-rojas-698a877b/

scholar.google.com/citations?user=WdqdQyYAAAAJ&hl=es

github.com/GaboTalero/